Faculty of Engineering & Technology Electrical & Computer Engineering Department

Circuits and Electronics Laboratory - ENEE2103

PreLab Exp7: BJT Transistor As An Amplifier, CE, CC, CB Connection

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Section: 3
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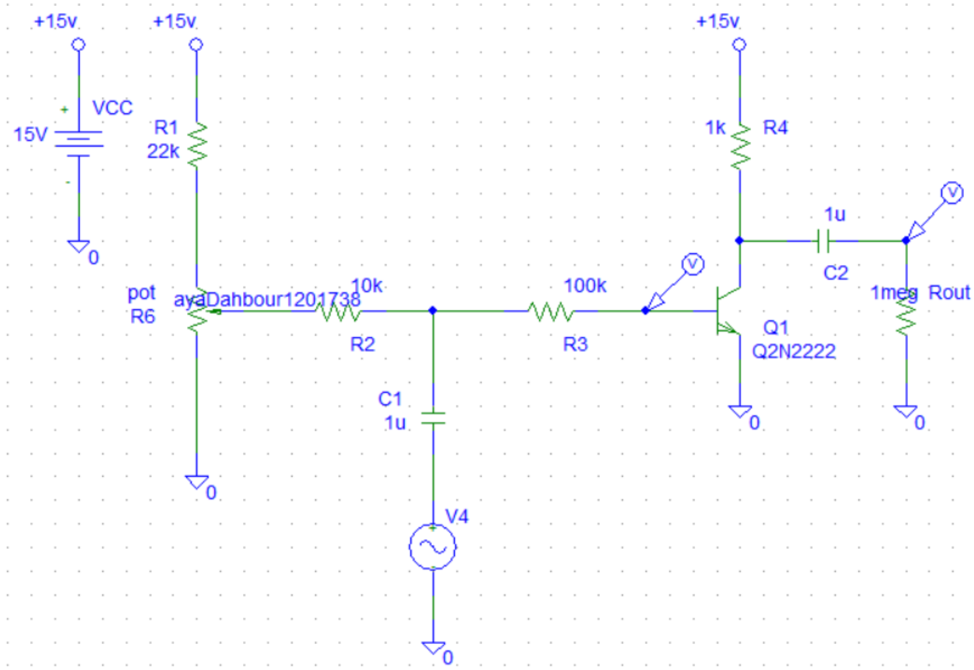
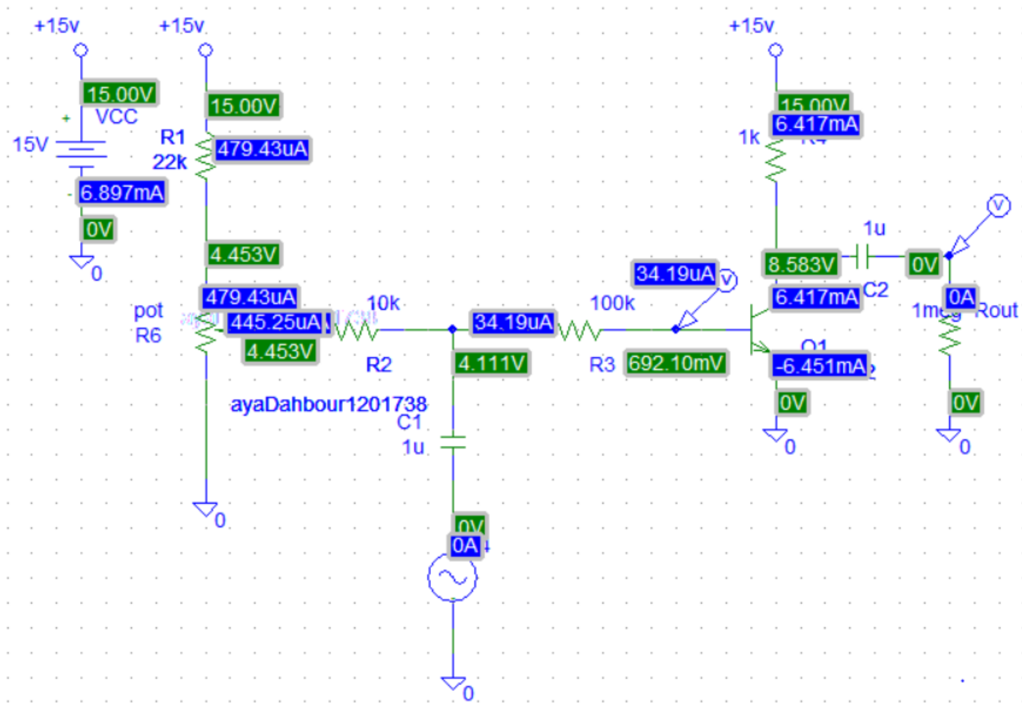
Part 1 :

The circuit connected in Pspice as follows :

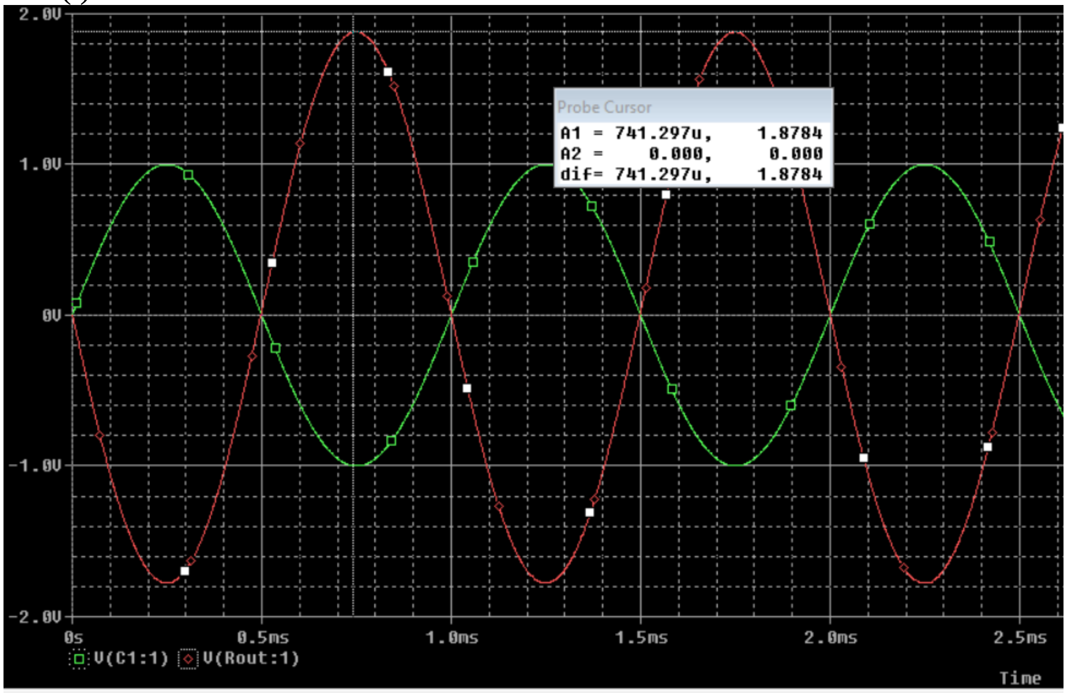


Then set Vi(t) amplitude to 0, set the potentiometer value to 10 k and its set value to 0. And sinusoidal source to 1 kHz and amplitude to zero.



V_C =8.5383v V_{BE}= 0.692v V_{CE} =8.583v I_C= 6.417mA I_B = 34.19uA

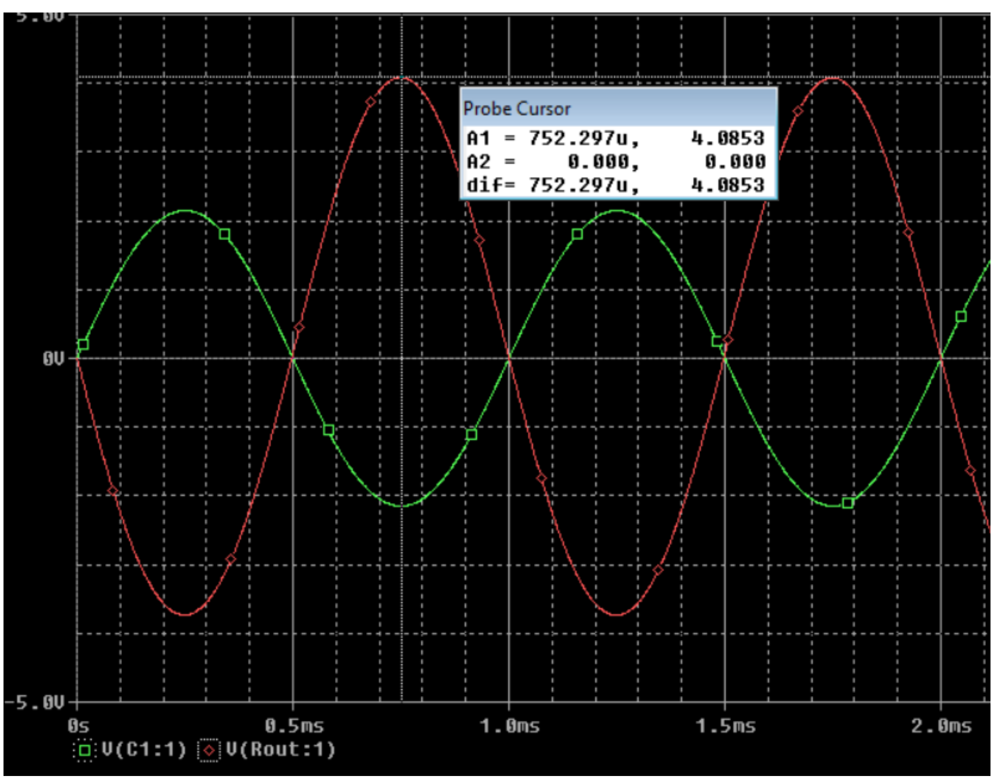
Then adjust amplitude of Vi(t) to 1 V
Vo(t) = 1.8784v



Change peak of Vi(t) such that Vo(t) =4V peak and perform Transient analysis

to find Vi : 1.8784 = 4 / Vi so Vi = 2.15v then adjust amplitude of Vi(t) to 2.15 V

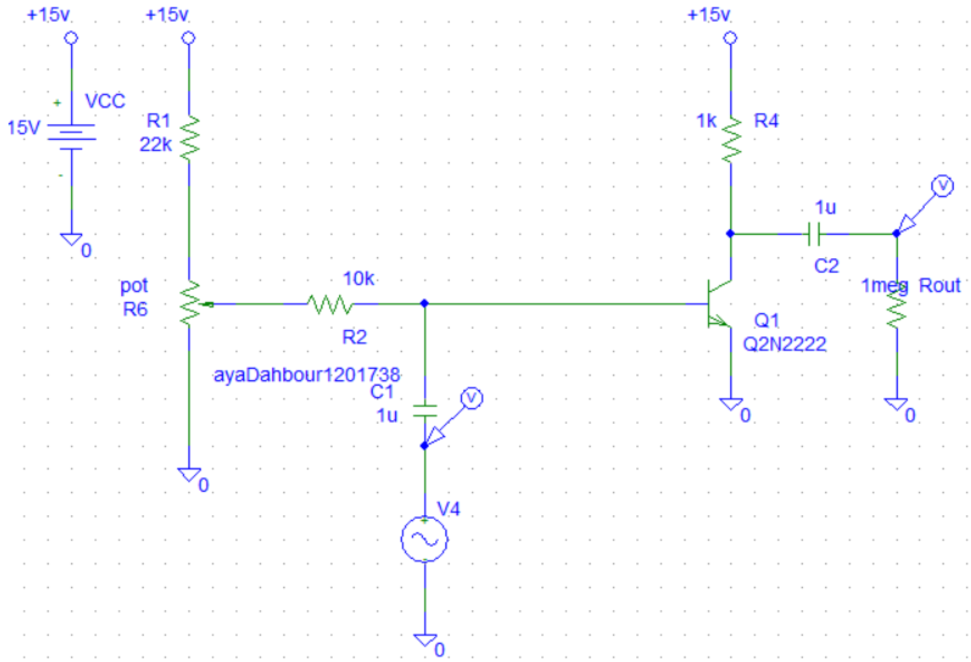
Vo = 4.085v

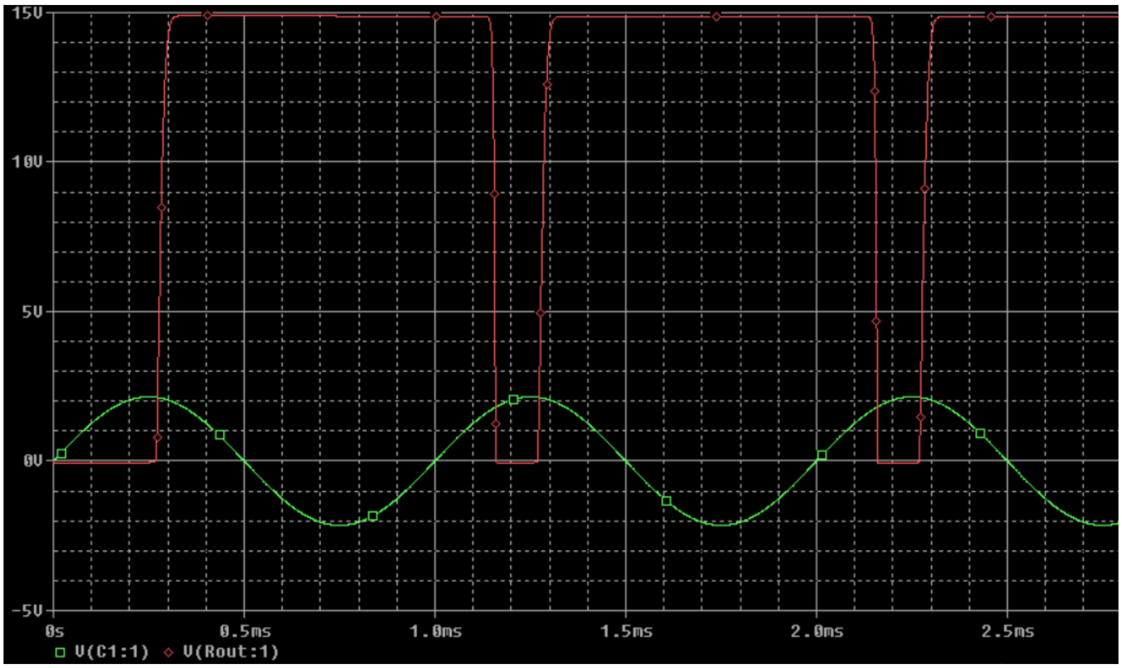


the voltage gain of the transistor Av = Vo (t) / Vi (t) = 4.085 / 2.15 = 1.9

Av_1 = Vo (t) / V_B (t) = 4.085 / 0.692 = 5.9

** Remove the 100k resistor and see what happens to voltage gain :



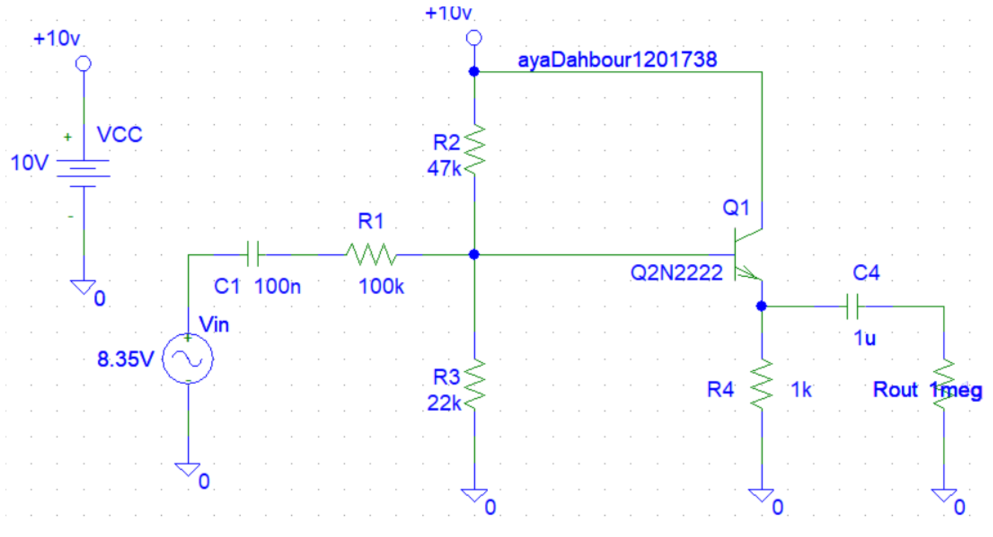


Av = Vo (t) / Vi (t) = 15/2.15 = 6.976 v

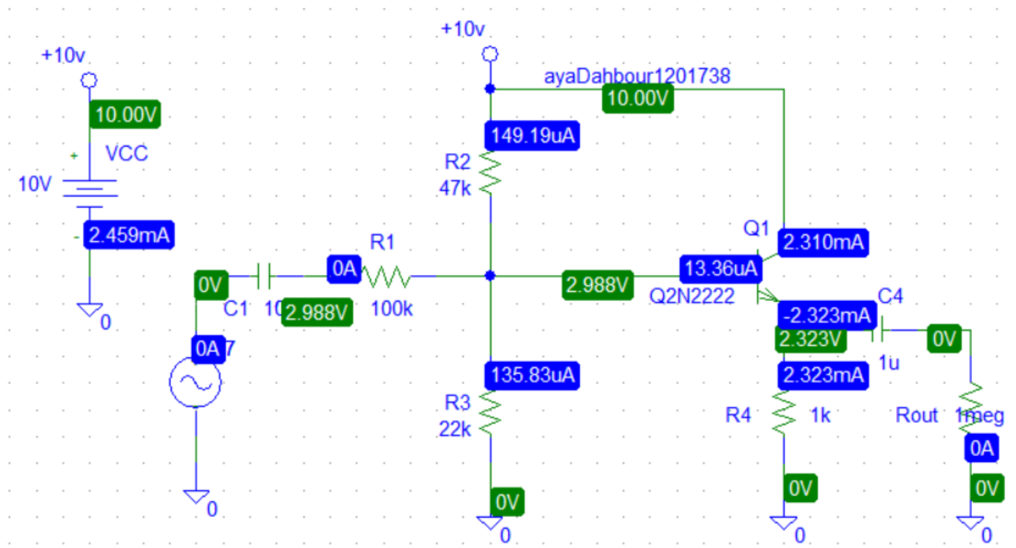
Part 2 :

COMMON COLLECTER TRANSISTOR AMPLIFIER

The circuit connected in Pspice as follows :

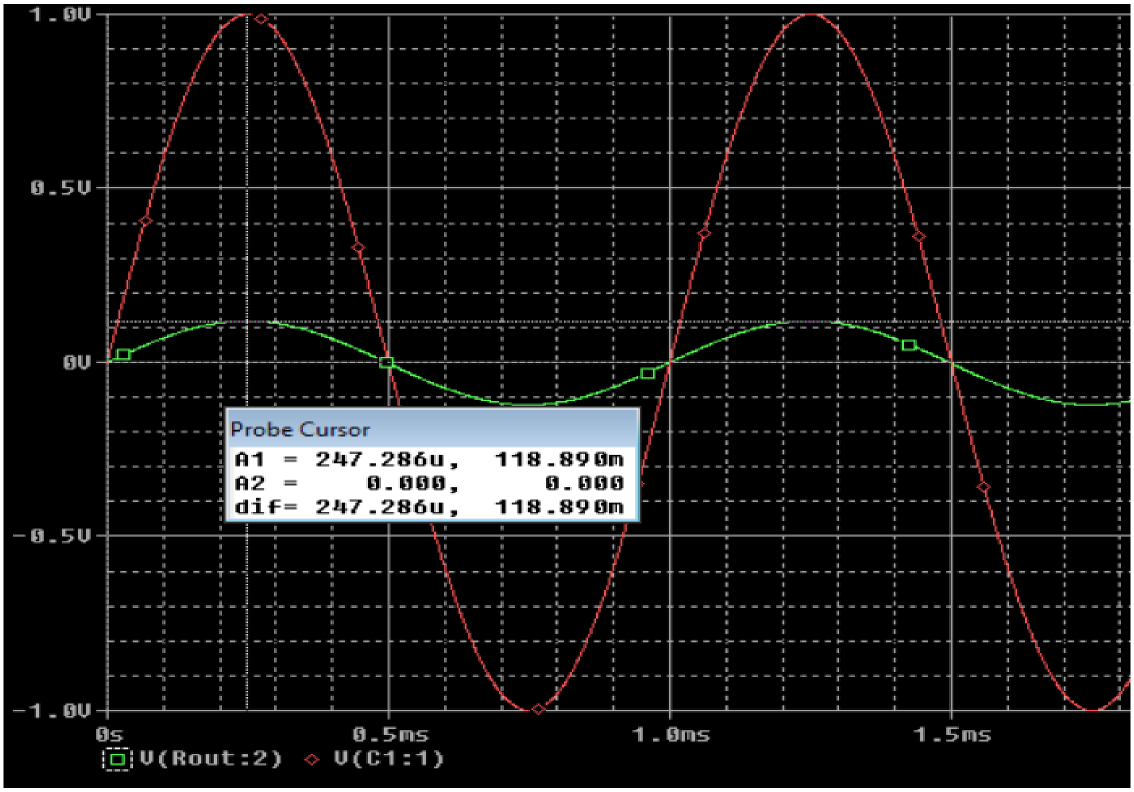


Set the sine wave generator to a frequency of 1 kHz , and its output amplitude to zero

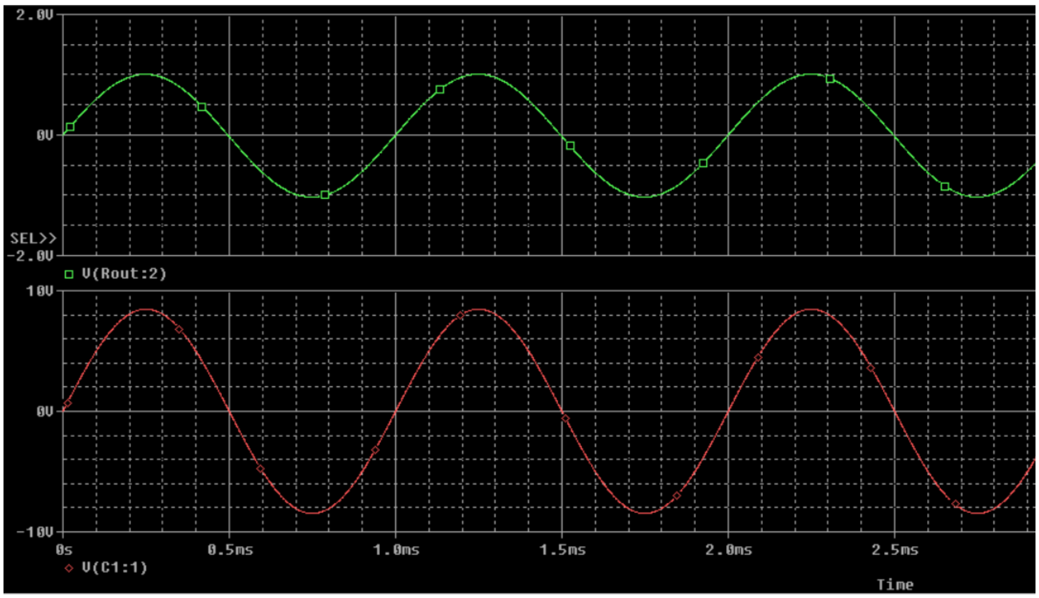


VB= 2.988v VC=10v IB=13.36uA IC=2.31mA

Adjust the amplitude of the sine wave generator until an output amplitude from the amplifier is about 2 volts peak-to-peak.

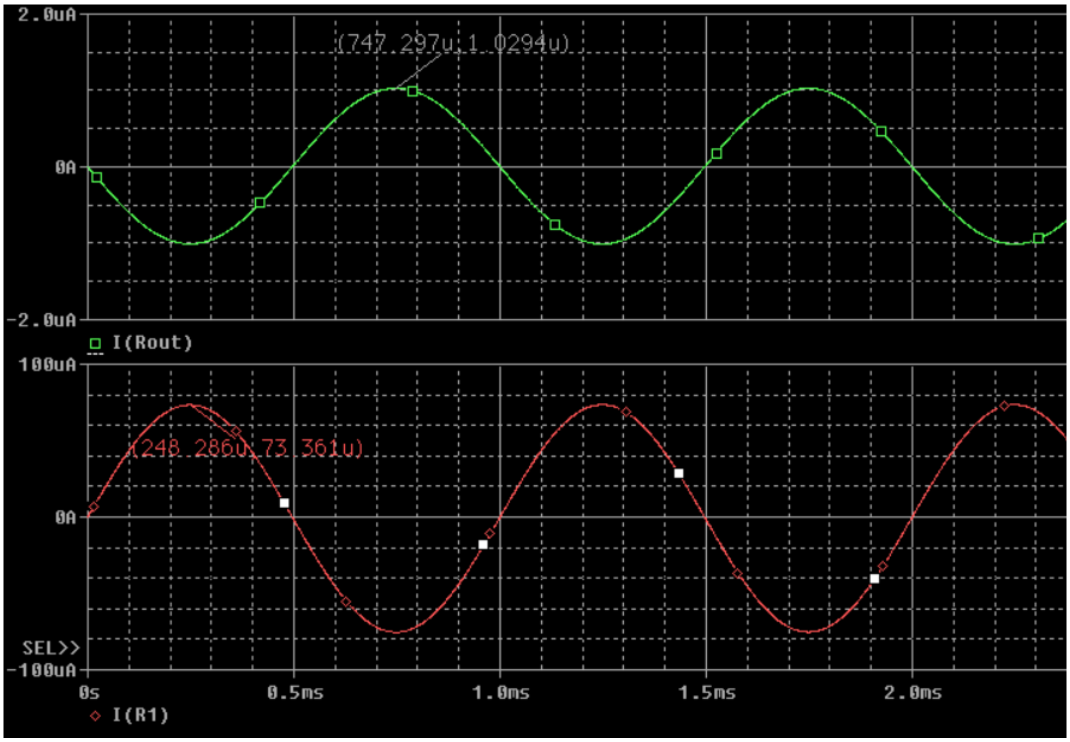


>>> 1v peak , To get 1v : Vi = 1/0.118 = 8.47v



the voltage gain Av = Vo/Vi = 1/8.47 = 8.47

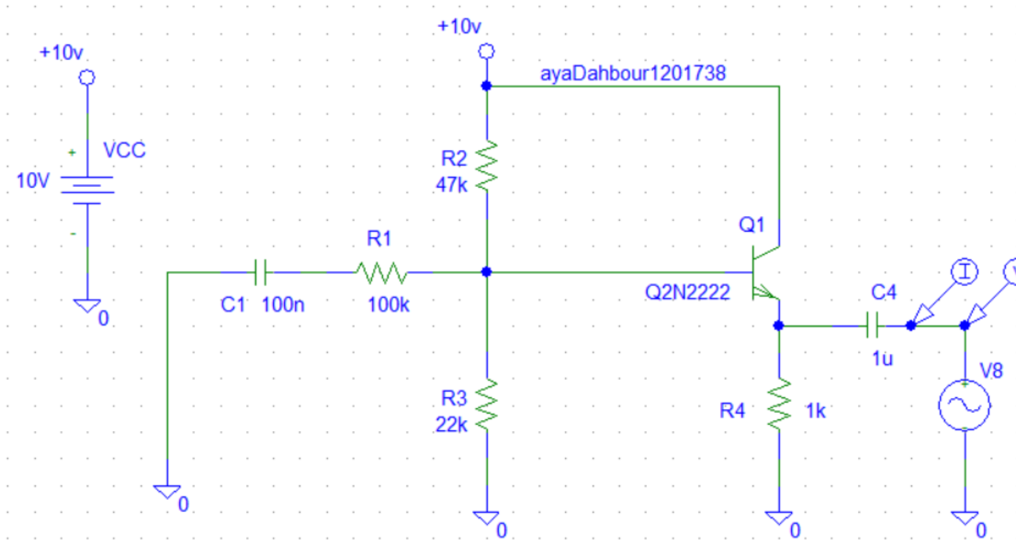
The input and output currents are displayed as shown below :



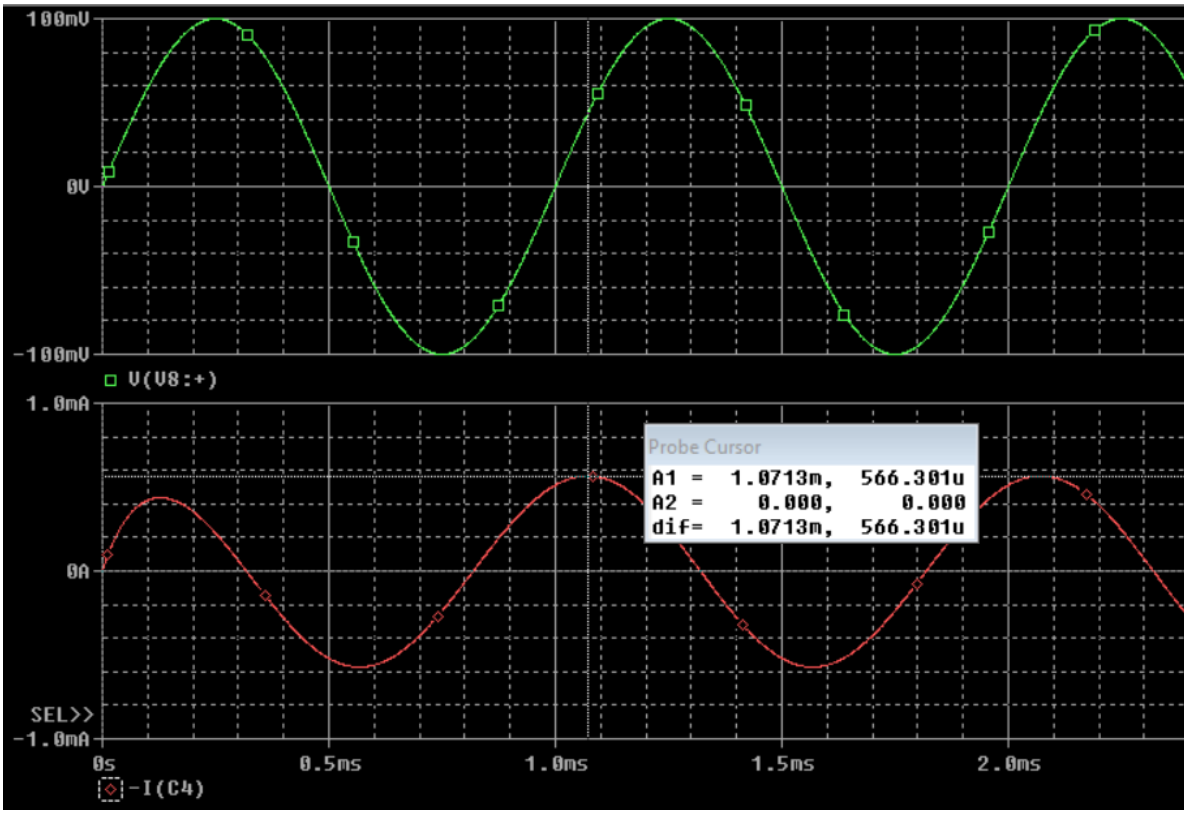
Iin = 73.361uA amplitude, Io = 1.074uA, so the current gain Ai= Io/Iin = 1.074/73.361 = 14.63 m

Zi = Vin / Iin = 8.47/73.36u = 115458 Ω

Finally find the output impedance of the amplifier :



The plots of the current and voltage are both displayed in the figure below as the voltage is in green and the current is in red.



Zo = Vo/Vin = 100m/566.3u = 176.5 Ω

Quantity	**Measured values**
V_{in}	**8.47v**
V_{out}	**1v**
V_{100k_RMS}	**7.2929v**
i_{out}	**1.074uA**
	Calculated values
$A_V=V_{out}/V_{in}$	**0.118**
$i_{in} = V_{100k\_RMS}/100k$	**72.929uA**
$A_i=i_{out}/i_{in}$	**14.36m**
$Z_{in}=V_{in}/i_{in}$	**115458**
$Z_{out}=V_T/i_T$	**176.5**